# CLR as attention: compendium in, module map out, no learned parameters

**WORKED EXAMPLE (TOY): not results**

**Input: expression compendium**

genes × experiments

toy: 14 x 60; real: thousands of genes, hundreds+ experiments, many labs

1 **Dependence** MI(i, j) → 2 **CLR** per-gene z-score → 3 **Softmax** per-gene temp. → 4 **Walk** t steps + residual

**no learned parameters | unsupervised | de novo**

**Output: module map**

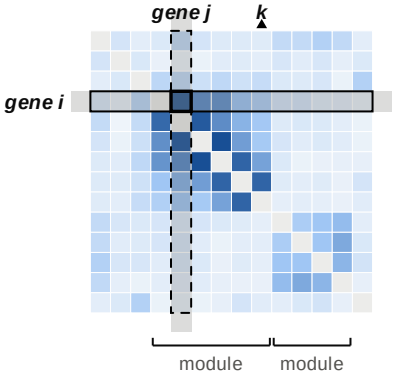
$K_9$ (toy) gene x gene map

outlined: modules, e.g. regulons, complexes, operons (generic types)

## 1 Dependence

*CLR (2007)*

pairwise compatibility (parameter-free, symmetric)



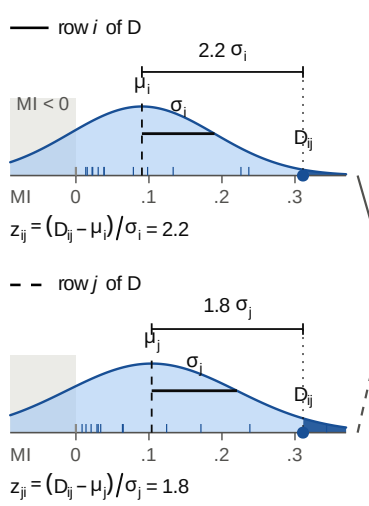

$D_{ij} = MI(i, j) = 0.31$

swap-in slot: any dependence measure (MI, rank correlation)

## 2 Per-gene calibration (CLR)

*CLR (2007)*

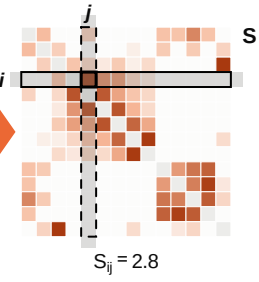
per-query normalization of the logits




$S_{ij} = \frac{z_{ij}^{+} + z_{ji}^{+}}{\sqrt{2}}$

$(2.2 + 1.8)/\sqrt{2} = 2.8$

$z^{+} = \max(z, 0)$




$S_{ij} = 2.8$

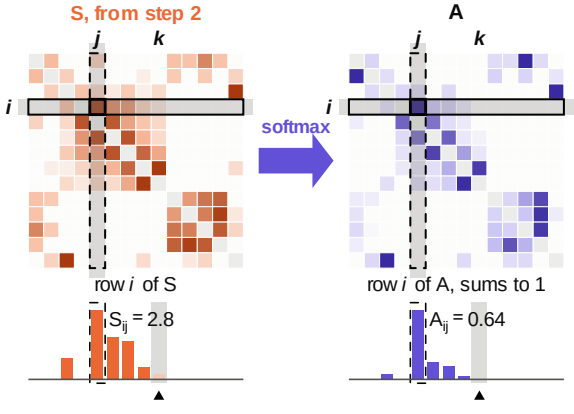
white = clipped to 0

**S is the input of step 3**

## 3 Softmax attention

*added here*

softmax over the keys, temperature per query




dark: top 5 kept (toy cap)
light: positive but cut

*k* (triangle): $S_{ik} > 0$ but 6th, cut by cap: $A_{ik} = 0$

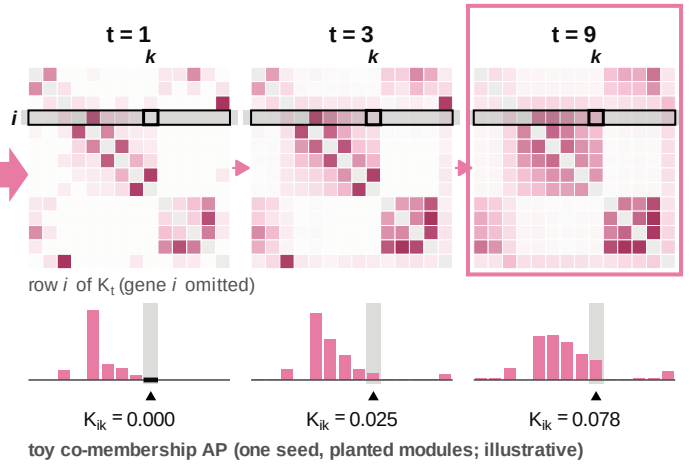
$A_i = \text{softmax}(S_i/\tau_i)$; $\tau_i$ set so that $k_{eff} = 3$

(toy $k_{eff}$ 3, cap 5; real method: $k_{eff}$ 10, top 50)

## 4 Walk: depth

*added here*

repeated attention with residual (cf. attention rollout)




**toy co-membership AP (one seed, planted modules; illustrative)**

0.69 — 0.77 — 0.91

orange tick: plain CLR S (0.77); grey tick: random (0.23)

$P = 0.5I + 0.5A$, $K_t = (P^t + (P^t)^T)/2$

magenta frame: t = 9, the depth used (chosen without labels).
0.5 I: residual term. Grey slot, triangle: gene *k*, $A_{ik} = 0$.
No value step: the readout is the attention map $K_t$ itself.

**No learned parameters: every step is computed from the data. Analogy with transformer attention, not equivalence.**

Toy: 14 genes, 60 experiments, planted modules; $k_{eff} = 3$, top 5, t = 1, 3, 9. Real method: $k_{eff} = 10$, top 50, t = 9.

Diagonal (self) set to 0 and kept among the values giving mu and sigma, as in CLR; drawn grey. Normal fit extends below MI = 0.

Light = low, dark = high within each matrix. Strip and examples use the toy only; no real-data result is shown.